

LE1_Review_Questions-Sound_and_Hearing

October 8, 2026

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[1]: ## Setup!  
import math
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5. The bulk modulus of a fluid were somehow quadrupled while its density stayed the same. By what factor would the speed of sound in the fluid change?

$$v = \sqrt{\frac{4B}{\rho}} \quad (1)$$

$$= 2\sqrt{\frac{B}{\rho}} \quad (2)$$

6. The pressure amplitude of a sound wave is doubled. What happens to the intensity and to the sound intensity level?

$$I = \frac{2p_{max}^2}{2\rho v} \quad (3)$$

$$(4)$$

7. Two (2) identical machines each produce a sound level of $\beta = 70$ dB at a listener's position. What is the approximate level when both operate together?
8. Which statement about the ends of a pipe containing a standing sound wave is correct?
- 9.
14. A sound wave in air has a pressure amplitude $p_{max} = 3.0$ Pa. Use $\rho = 1.20 \frac{\text{kg}}{\text{m}^3}$ and wave speed $v = 344 \frac{\text{m}}{\text{s}}$. What is the intensity of the wave?

$$I = \frac{p_{max}^2}{2\rho v} \quad (5)$$

$$= \frac{(3.0 \text{ Pa})^2}{2 \times 1.20 \frac{\text{kg}}{\text{m}^3} \times 344 \frac{\text{m}}{\text{s}}} \quad (6)$$

$$= \frac{9.0 \text{ Pa}^2}{825.6 \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}} \quad (7)$$

$$= 0.0109 \frac{\text{W}}{\text{m}^2} \quad (8)$$

$$= 1.09 \times 10^{-2} \frac{\text{W}}{\text{m}^2} \quad (9)$$

$$(10)$$

15. For the same wave as above, what is the sound intensity level?

$$\beta = (10 \text{ dB}) \log_{10}\left(\frac{I}{I_0}\right) \quad (11)$$

$$= (10 \text{ dB}) \log_{10}\left(\frac{1.09 \times 10^{-2} \frac{\text{W}}{\text{m}^2}}{10^{-12} \frac{\text{W}}{\text{m}^2}}\right) \quad (12)$$

$$= 10 \text{ dB} \times 10 \quad (13)$$

$$= 100 \text{ dB} \quad (14)$$

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[2]: p_max = 3.0  ## Pa
rho   = 1.20  ## kg / m^3
v     = 344.0  ## m / s
I     = p_max**2 / (2 * rho * v)  ## W / m^2
display(I)

I0 = 10**-12  ## W / m^2
beta = 10 * math.log10(I / I0)
display(math.log10(I / I0))
display(beta)
```

0.010901162790697675

10.037472825156188

100.37472825156188

17. The speed of sound in air at 20 degrees C is $v = 344 \frac{\text{m}}{\text{s}}$. What is it at 0 degrees C? Treat air as an ideal gas.

First convert degrees C to degrees K.

$$20^{\circ}\text{C} + 273.15^{\circ}\text{K} = 293.15^{\circ}\text{K} \quad (15)$$

$$0^{\circ}\text{C} + 273.15^{\circ}\text{K} = 273.15^{\circ}\text{K} \quad (16)$$

$$(17)$$

Then

$$v = \sqrt{\frac{\gamma RT}{M}} \quad (18)$$

$$(19)$$

18.

[3]: 20+273.15

[3]: 293.15

18. An organ pipe has a length $L = 0.60$ m. Use wave speed $v = 344 \frac{\text{m}}{\text{s}}$. What are the fundamental frequencies if the pipe is (i) open at both ends, and (ii) closed at one end?

For an open pipe:

$$f_1 = \frac{v}{2L} \quad (20)$$

$$= \frac{344 \frac{\text{m}}{\text{s}}}{2 \times 0.60 \text{ m}} \quad (21)$$

$$= 287 \frac{1}{\text{Hz}} \quad (22)$$

$$= 287 \text{ Hz} \quad (23)$$

For a closed pipe:

$$f_1 = \frac{v}{4L} \quad (24)$$

$$= \frac{344 \frac{\text{m}}{\text{s}}}{4 \times 0.60 \text{ m}} \quad (25)$$

$$(26)$$

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[4]: L = 0.60 ## m
v = 344.0 ## m / s
display(L, v)
```

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f_1 = v / (2 * L)
display(f_1)
```

```
f_1 = v / (4 * L)
display(f_1)
```

0.6

344.0

286.6666666666667

143.33333333333334

19. A pipe closed at one end has a fundamental frequency $f_1 = 220$ Hz with wave speed $v = 344 \frac{\text{m}}{\text{s}}$. What is its length, and which of 440 Hz and 660 Hz can it produce?

$$f_1 = \frac{v}{4L} \quad (27)$$

$$4Lf_1 = v \quad (28)$$

$$L = \frac{v}{4f_1} \quad (29)$$

$$= \frac{344 \frac{\text{m}}{\text{s}}}{4 \times 220 \frac{1}{\text{s}}} \quad (30)$$

$$= 0.391 \text{ m} \quad (31)$$

```
[5]: f_1 = 220.0 ## Hz
v = 344.0 ## m / s
display(v, f_1)

L = v / (4 * f_1)
display(L)
display(344 / (4 * 220))

for i in range(1, 10, 2):
    display(f_1 * i)
```

344.0

220.0

0.39090909090909093

0.39090909090909093

220.0

660.0

1100.0

1540.0

1980.0

23. A listener walks at $v_L = 20 \frac{\text{m}}{\text{s}}$ toward a stationary source (i.e. $v_S = 0 \frac{\text{m}}{\text{s}}$) that emits $f_S = 500 \text{ Hz}$ with a wave speed $v = 344 \frac{\text{m}}{\text{s}}$. What frequency does the listener hear?

$$f_L = \frac{v + v_L}{v + v_S} f_S \quad (32)$$

$$= \frac{344 \frac{\text{m}}{\text{s}} + 20 \frac{\text{m}}{\text{s}}}{344 \frac{\text{m}}{\text{s}} + 0 \frac{\text{m}}{\text{s}}} \times 500 \text{ Hz} \quad (33)$$

$$= \frac{364 \frac{\text{m}}{\text{s}}}{344 \frac{\text{m}}{\text{s}}} \times 500 \text{ Hz} \quad (34)$$

```
[6]: v_L = 20.0
v_S = 0.0
f_S = 500.0
v = 344.0
display(v_L, v_S, f_S, v)

f_L = (v + v_L) / (v + v_S) * f_S
display(f_L)
```

20.0

0.0

500.0

344.0

529.0697674418606